



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2014

**HIGHER SCHOOL
CERTIFICATE
ASSESSMENT TASK #3**

Mathematics Extension 1

General Instruction

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen.
Pencil may be used for diagrams.
- Board approved calculators maybe used.
- Start each **NEW** question in a separate answer booklet.
- Marks may **NOT** be awarded for messy or badly arranged work.
- All necessary working should be shown in every question if full marks are to be awarded
- Answer in simplest exact form unless otherwise instructed.
- A table of standard integrals is provided
- There are 4 questions

Total Marks – 75

Question 1 – **20 Marks**

Question 2 – **20 Marks**

Question 3 – **15 Marks**

Question 4 – **20 Marks**

Attempt all questions

Examiner: *R. Boros*

This is an assessment task only and does not necessarily reflect
the content or format of the Higher School Certificate

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

Question One (20 marks)

Marks

Use a SEPARATE answer booklet.

- (a) Differentiate with respect to x , $\tan^{-1} 3x$. 2

- (b) Find 2

$$\int x\sqrt{x^2 + 2} \cdot dx$$

using the substitution $u = x^2 + 2$.

- (c) (i) Draw a neat sketch of $y = \sin^{-1} x$. 1

- (ii) State the domain and range. 2

- (iii) Using your sketch in Questions 1 (c) (i), shade the region bounded by the curve $y = \sin^{-1} x$, the x axis and the line $x = 1$. Using Simpson's Rule with 3 function values, find the approximation to this shaded area correct to 2 decimal places. 4

- (d) Given that 3

$$\int_0^1 \frac{dx}{x^2 + 3} = A\pi$$

find the exact value of A .

- (e) A particle moves on a line so that its distance x from the origin at time t is given by $x = t^3 - 9t^2 + 15t - 7$. Find when and where the particle first comes to rest. 3

- (f) Find the coordinates of the vertex and the focus and the equation of the directrix given the parabola $y = x^2 - 4x$. 3

End of Question One

Question Two (20 marks)

Marks

Use a SEPARATE answer booklet.

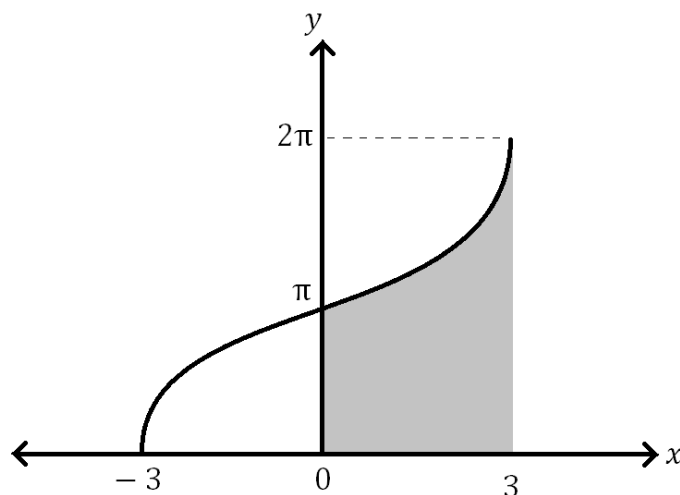
- (a) The rate at which a body cools in air is assumed to be proportional to the difference between its temperature T and the constant temperature S of the surrounding air. This can be expressed by the differential equation

$$\frac{dT}{dt} = k(T - S)$$

where t is the time in hours and k is a constant.

- | | | |
|-------|---|---|
| (i) | Show that $T = S + Be^{kt}$, where B is a constant is a solution of the differential equation. | 2 |
| (ii) | A heated meal cools from 80°C to 40°C in 2 hours. The air temperature S in the room where the meal is placed is 20°C . Find the value of k correct to 4 decimal places. | 2 |
| (iii) | Find the temperature of the meal, correct to the nearest degree after 1 further hour has elapsed. | 1 |

- (b) The graph below is the function $y = a \cos^{-1} bx$ where a and b are constants.



- | | | |
|------|---|---|
| (i) | Find the values of a and b . | 2 |
| (ii) | Hence find the exact value of the shaded area. | 4 |
| (c) | A particle moves along the x axis with acceleration $\frac{d^2x}{dt^2} = 70 + 12t - 12t^2$. If the particle is initially at rest at the origin, find its maximum displacement in the positive direction. | 4 |

Question Two continues on next page

- (d) A particle undergoes Simple Harmonic Motion about the origin O. Its displacement x cm from O at time t seconds is given by

$$x = 3 \cos \left(2t + \frac{\pi}{3} \right)$$

- | | | |
|-------|--|---|
| (i) | Express the acceleration as a function of the displacement. | 2 |
| (ii) | What is the amplitude of the motion? | 1 |
| (iii) | Find the value of x for which the speed is a maximum and determine this speed. | 2 |

End of Question Two

Question Three (15 marks)

Marks

Use a SEPARATE answer booklet.

- (a) A spherical bubble is expanding so that its volume is increasing at the constant rate of 10mm^3 per second. 3

What is the rate of increase of the radius when the surface area is 500mm^2 .

- (b) (i) Differentiate with respect to x , 2

$$x \sin^{-1} x + \sqrt{1 - x^2}$$

- (ii) Hence, evaluate 2

$$\int_0^1 \sin^{-1} x \cdot dx$$

leaving your answer in exact form.

- (c) Use the substitution $u = x^3 + 3x - 2$ to evaluate 3

$$\int_0^1 (x^2 + 1) \sqrt[3]{(x^3 + 3x - 2)^5} \cdot dx$$

- (d) A particle moves on a line so that its distance from the origin O at time t is x .

- (i) Given $\frac{d^2x}{dt^2} = 10x - 2x^3$ and $v = 0$ at $x = 1$, find v in terms of x . 3

- (ii) Is the motion Simple Harmonic? Explain. 2

End of Question Three

Question Four (20 marks)

Marks

Use a SEPARATE answer booklet.

(a)

(i) Show that $\tan^{-1} x + \tan^{-1} y = \tan^{-1} \left(\frac{x+y}{1-xy} \right)$

2

(ii) Hence or otherwise, find the positive value of x satisfying the equation

3

$$\tan^{-1}(2x) + \tan^{-1}(3x) = \frac{\pi}{4}$$

(b) A particle executes Simple Harmonic Motion with period T seconds and amplitude A cm. Find its maximum velocity in terms of A and T .

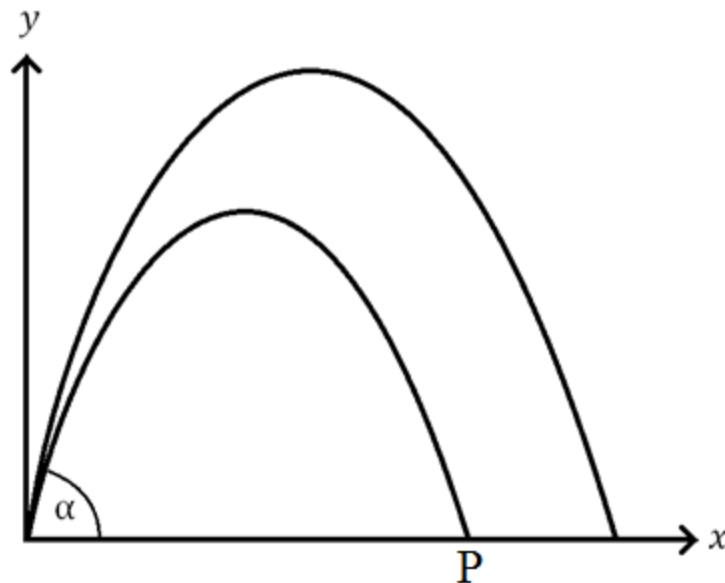
3

(c) A cricket ball and a tennis ball are thrown simultaneously from the same point and in the same direction and with the same non-zero angle of projection (upward inclination to the horizontal), α but with different velocities U and V metres per second, where $U < V$.

5

The slower tennis ball hits the ground first at a point P on the same level as the point of projection.

Show that, while the 2 balls are in flight, the line joining them has an inclination to the horizontal which is independent of time.



Question Four continues on next page

- (d) Tangents are drawn from the external point $P(x_0, y_0)$ to the parabola $x^2 = 4y$. These tangents touch the parabola at Q and R respectively.

- (i) Prove that the midpoint T of QR is 5

$$\left(x_0, \frac{1}{2}x_0^2 - y_0\right)$$

- (ii) If P moves on the line $x - y = 1$, find the equation of the locus of T. 2

End of Question Four
End of Exam

SBHS 2014 Extension 1 Assessment #3 Solutions

QUESTION ONE.

$$(a) \quad y = \tan^{-1} 3x$$

$$y' = \left| \frac{3}{1+9x^2} \right|$$

$$(b.) \quad \text{Let } u = x^2 + 2$$

$$du = 2x \, dx.$$

$$\therefore I = \frac{1}{2} \int \sqrt{u} \cdot du$$

$$= \frac{1}{3} u^{3/2} + C$$

$$= \left| \frac{1}{3} (x^2 + 2)^{3/2} + C \right|$$

$$(c) \quad (i) \quad \left(\begin{array}{c} \text{Graph of } y = \tan^{-1} x \text{ on the interval } [-1, 1]. \\ \text{The curve passes through } (-1, -\frac{\pi}{4}), (0, 0), \text{ and } (1, \frac{\pi}{4}). \end{array} \right)$$

$$(ii) \quad \left(\begin{array}{l} D: |x| \leq 1. \\ R: |\tan^{-1} x| \leq \frac{\pi}{2}. \end{array} \right)$$

$$(iii) \quad A \doteq \frac{\frac{1}{2}}{3} \left[f(0) + f(1) + 4f\left(\frac{1}{2}\right) \right]$$

$$= \frac{1}{6} \left[0 + \frac{\pi}{2} + 4 \times \frac{\pi}{6} \right]$$

$$= \frac{1}{6} \left(\frac{\pi}{2} + \frac{2\pi}{3} \right)$$

$$= \frac{\frac{7\pi}{6}}{36}$$

$$\doteq \left| 0.61 \, u^2 \right|$$

$$(d) \int_0^1 \frac{dx}{x^2+3} = A\pi.$$

$$LHS = \frac{1}{\sqrt{3}} \left[\tan^{-1} \frac{x}{\sqrt{3}} \right]_0^1$$

$$= \frac{1}{\sqrt{3}} \tan^{-1} \frac{1}{\sqrt{3}}.$$

$$= \frac{1}{\sqrt{3}} \cdot \frac{\pi}{6}.$$

$$\therefore \frac{\pi}{6\sqrt{3}} = A\pi.$$

$$\therefore A = \frac{1}{6\sqrt{3}}$$

$$= \left| \frac{\sqrt{3}}{18} \right|$$

$$(e) x = t^3 - 9t^2 + 15t - 7.$$

$$\dot{x} = 3t^2 - 18t + 15.$$

$$= 3(t^2 - 6t + 5)$$

$$= 3(t-1)(t-5)$$

$\therefore$ At rest when $\dot{x} = 0$

$$t = 1, 5.$$

$\therefore$ First time is after 1 unit of time
at $x = 0$.

$$(f) y = x^2 - 4x$$

$$\therefore (x-2)^2 = y+4$$

$$(x-2)^2 = 4 \times \frac{1}{4} (y+4)$$

Vertex is $(2, -4)$

$\therefore$ Focus $(2, -3\frac{3}{4})$

Directrix $|y = -4\frac{1}{4}|$

Question 2

(a) (i) $T = S + Be^{kt} \quad \therefore Be^{kt} = T - S$

$$\begin{aligned}\frac{dT}{dt} &= kBe^{kt} \\ &= k(T - S)\end{aligned}$$

$\therefore$ It is a solution to the DE

(ii) $S = 20$

Now $80 = 20 + Be^{k \cdot 0}$ when $t = 0$

$$\therefore B = 60$$

When $t = 2$, $T = 40$

$$40 = 20 + 60e^{2k}$$

$$20 = 60e^{2k}$$

$$e^{2k} = \frac{1}{3}$$

$$2k = \ln \frac{1}{3}$$

$$k = \frac{1}{2} \ln \frac{1}{3}$$

$$\approx -0.5493$$

(iii) When $t = 3$

$$T = 20 + 60e^{3k}$$

$$\approx 31.55^0$$

(b) (i) $y = a \cos^{-1} bx$

When $x = 0$, $y = \pi$

$$\pi = a \cos^{-1} b \times 0$$

$$\pi = a \frac{\pi}{2}$$

Thus $a = 2$

When $x = 3$, $y = 2\pi$

$$2\pi = 2 \cos^{-1} 3b$$

$$\cos \pi = 3b$$

Thus $b = -\frac{1}{3}$

$$\therefore y = 2 \cos^{-1} \left(-\frac{x}{3} \right)$$

$$(ii) \quad \frac{y}{2} = \cos^{-1}\left(-\frac{x}{3}\right)$$

$$-\frac{x}{3} = \cos\left(\frac{y}{2}\right)$$

$$x = -3\cos\left(\frac{y}{2}\right)$$

Area consists of rectangle plus area between line $x = 3$ and curve.

$$\begin{aligned} A &= 3\pi + \int_{\pi}^{2\pi} \left(3 - \left(-3\cos\left(\frac{y}{2}\right) \right) \right) dy \\ &= 3\pi + \left[3y + 3 \times 2 \sin\left(\frac{y}{2}\right) \right]_{\pi}^{2\pi} \\ &= 3\pi + [6\pi + 6\sin\pi] - \left[3\pi + 6\sin\left(\frac{\pi}{2}\right) \right] \\ &= 6\pi - 6 \text{ sq units} \end{aligned}$$

$$(c) \quad \frac{d^2y}{dx^2} = 70 + 12t - 12t^2$$

When $t = 0$, $x = 0$, $v = 0$.

$$\frac{dy}{dx} = 70t + 6t^2 - 4t^3 + C, \quad C = 0$$

$$\therefore v = 70t + 6t^2 - 4t^3$$

$$\text{Hence } x = 35t^2 + 2t^3 - t^4 + D, \quad D = 0$$

$$\therefore x = 35t^2 + 2t^3 - t^4$$

Max displacement when $v = 0$.

$$0 = 70t + 6t^2 - 4t^3$$

$$0 = 2t(35 + 3t - 2t^2)$$

$$t = 0, \frac{3 \pm 17}{4}$$

$$t = -\frac{7}{2}, 0, 5$$

Clearly $t > 0$, so max x is when $t = 5$.

$$x_{\max} = 35(5)^2 + 2(5)^3 - (5)^4 = 500$$

(d) $x = 3\cos\left(2t + \frac{\pi}{2}\right)$

(i) $\dot{x} = -6\sin\left(2t + \frac{\pi}{2}\right)$

$$\ddot{x} = -12\cos\left(2t + \frac{\pi}{2}\right)$$

$$\therefore \ddot{x} = -4x$$

(ii) $a = 3$

(iii) Max speed when $\ddot{x} = 0$

$$0 = -12\cos\left(2t + \frac{\pi}{2}\right)$$

$$\therefore \cos\left(2t + \frac{\pi}{2}\right) = 0$$

$$2t + \frac{\pi}{2} = \frac{\pi}{2}$$

$$\therefore t = 0$$

At this time

$$\dot{x} = -6\sin\left(\frac{\pi}{2}\right)$$

$$= -6$$

Hence max speed = 6 cm/s.

Q.3.

Extension 1 - 15 Marks

(a) $\frac{dV}{dt} = 10 \text{ mm}^3/\text{s}$ Find $\frac{dr}{dt}$ when $S = 500 \text{ mm}^2$

Now $V = \frac{4}{3} \pi r^3$

$\Rightarrow S = \frac{dV}{dr} = 4\pi r^2$

Then $\frac{dV}{dr} = \frac{dV}{dt} \frac{dt}{dr}$

$500 = 10 \times \frac{dt}{dr}$

$\frac{dt}{dr} = 50$

$\Rightarrow \frac{dr}{dt} = \frac{1}{50} \text{ mm/s}$

(3)

(b) (i) $y = x \sin^{-1} x + \sqrt{1-x^2}$

$y' = x \cdot \frac{1}{\sqrt{1-x^2}} + \sin^{-1} x \cdot 1 + \frac{1}{2} (1-x^2)^{-1/2} (-2x)$

$y' = \frac{x}{\sqrt{1-x^2}} - \frac{x}{\sqrt{1-x^2}} + \sin^{-1} x$

$\Rightarrow y' = \sin^{-1} x$

(ii) $\therefore \int_0^1 \sin^{-1} x \, dx = \left[x \sin^{-1} x + \sqrt{1-x^2} \right]_0^1$

$= (\sin^{-1} 1 + 0) - (0 + 1)$

$= \sin^{-1} 1 - 1$

$= \frac{\pi}{2} - 1$

(2)

$$3(c) \int_0^1 (x^2+1) \cdot \sqrt[3]{(x^3+3x-2)^5} dx$$

$$\text{let } u = x^3 + 3x - 2 \\ du = 3(x^2+1)dx$$

$$I = \int (x^2+1) u^{5/3} \frac{du}{3(x^2+1)} \quad \checkmark$$

$$= \frac{1}{3} \int u^{5/3} du$$

$$= \frac{1}{3} u^{8/3} \times \frac{3}{8} \quad \checkmark$$

$$= \frac{1}{8} u^{8/3}$$

(3)

$$\text{Then Definite Int} = \frac{1}{8} \left[(x^3+3x-2)^{8/3} \right]_0^1$$

$$= \frac{1}{8} \left(2^{8/3} - (-2)^{8/3} \right)$$

$$= 0 \quad \checkmark$$

$$(d)(i) \ddot{x} = \frac{d(\frac{1}{2}v^2)}{dx} = 10x - 2x^3$$

$$\frac{1}{2}v^2 = 5x^2 - \frac{x^4}{2} + C \quad \checkmark$$

$$\text{When } v=0, x=1 \Rightarrow 0 = 5 - \frac{1}{2} + C$$

$$\Rightarrow C = -4\frac{1}{2} \quad \checkmark$$

$$\therefore \frac{1}{2}v^2 = 5x^2 - \frac{x^4}{2} - 4\frac{1}{2} \quad \checkmark$$

$$v^2 = 10x^2 - x^4 - 9 \quad \checkmark$$

(3)

$$(ii) \text{ Since } \ddot{x} = 10x - 2x^3$$

$$\neq -n^2x \quad \checkmark$$

$$\text{and } \neq -n^2(x-a) \quad \text{where } n \in \mathbb{Z}$$

$$\Rightarrow \underline{\text{NOT SHM.}} \quad \checkmark$$

(2)

Q4

(a) (i) RTP $\tan^{-1}x + \tan^{-1}y = \tan^{-1}\left(\frac{x+y}{1-xy}\right)$

$$\begin{aligned} & \tan(\tan^{-1}x + \tan^{-1}y) \\ &= \frac{\tan(\tan^{-1}x) + \tan(\tan^{-1}y)}{1 - \tan(\tan^{-1}x)\tan(\tan^{-1}y)} \\ &= \frac{x+y}{1-xy} \end{aligned}$$

$$\therefore \tan^{-1}x + \tan^{-1}y = \tan^{-1}\left(\frac{x+y}{1-xy}\right)$$

2

(ii) $\tan^{-1}(2x) + \tan^{-1}(3x) = \frac{\pi}{4}$

$$\therefore \tan^{-1}\left(\frac{2x+3x}{1-2x \cdot 3x}\right) = \frac{\pi}{4}$$

$$\therefore \frac{5x}{1-6x^2} = \tan \frac{\pi}{4} = 1$$

$$\therefore 5x = 1 - 6x^2$$

$$\therefore 6x^2 + 5x - 1 = 0$$

$$\therefore (6x-1)(x+1) = 0$$

$$\therefore x = \frac{1}{6} \text{ OR } x = -1.$$

$\therefore x > 0, x = \frac{1}{6}$

3

(b) Particle executes SHM

$$\therefore v^2 = n^2 (a^2 - x^2)$$

$$\text{As Period} = T = \frac{2\pi}{n}$$

$$n = \frac{2\pi}{T}$$

$$\therefore v^2 = \left(\frac{2\pi}{T}\right)^2 (A^2 - x^2)$$

Max velocity occurs when $x = 0$

$$\therefore v_{\max} = \frac{2\pi A}{T}$$

3

(c) Tennis ball:

$$\ddot{x} = 0$$

$$\dot{x} = U \cos \alpha$$

$$x = U \cos \alpha \cdot t$$

$$\ddot{y} = -g$$

$$\dot{y} = -gt + U \sin \alpha$$

$$y = -\frac{1}{2}gt^2 + U \sin \alpha \cdot t$$

Cricket ball

$$\ddot{x} = 0$$

$$\dot{x} = V \cos \alpha$$

$$x = V \cos \alpha \cdot t$$

$$\ddot{y} = -g$$

$$\dot{y} = -gt + V \sin \alpha$$

$$y = -\frac{1}{2}gt^2 + V \sin \alpha \cdot t$$

Gradient between balls

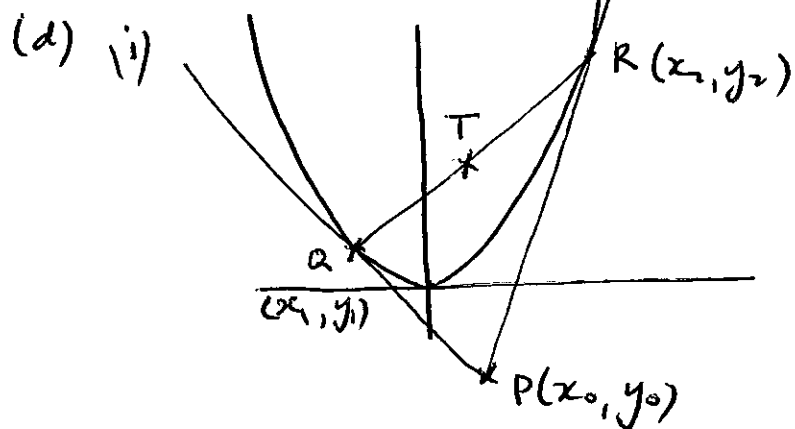
$$= \frac{-\frac{1}{2}gt^2 + V \sin \alpha \cdot t - (-\frac{1}{2}gt^2 + U \sin \alpha \cdot t)}{V \cos \alpha \cdot t - U \cos \alpha \cdot t}$$

$$= \frac{V \sin \alpha \cdot t - U \sin \alpha \cdot t}{V \cos \alpha \cdot t - U \cos \alpha \cdot t}$$

$$= \frac{(V-U) \sin \alpha \cdot t}{(V-U) \cos \alpha \cdot t}$$

$= \tan \alpha$ which is independent of t ,
 $\therefore$ Inclination to horizontal is independent of t

5



QR is $xx_0 = 2(y + y_0)$

For Q and R $xx_0 = 2\left(\frac{x^2}{2} + y_0\right)$

$$\therefore xx_0 = \frac{x^2}{2} + 2y_0$$

$$\therefore 2xx_0 = x^2 + 4y_0$$

$$\therefore x^2 - 2xx_0 + 4y_0 = 0$$

Sum of roots $x_1 + x_2 = 2x_0$

$$\therefore \frac{x_1 + x_2}{2} = x_0$$

$$\therefore x_T = x_0$$

As T lies on QR

$$x_T x_0 = 2(y_T + y_0)$$

$$\therefore x_0 x_0 = 2(y_T + y_0)$$

$$\therefore \frac{x_0^2}{2} = y_T + y_0$$

$$\therefore y_T = \frac{x_0^2}{2} - y_0$$

$$\therefore T \text{ is } \left(x_0, \frac{x_0^2}{2} - y_0\right)$$

5

(ii) As P moves on $x - y = 1$

$$x_0 - y_0 = 1$$

$$\therefore y_0 = x_0 - 1$$

$$\therefore y_T = \frac{x_0^2}{2} - y_0$$

$$\begin{aligned}\therefore y_T &= \frac{x_T^2}{2} - (x_0 - 1) \\ &= \frac{x_T^2}{2} - x_T + 1.\end{aligned}$$

$$\therefore \text{locus of T is } y = \frac{x^2}{2} - x + 1.$$

2